About me (in 2023-07-11)

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HTML: https://paulojeronimo.com/about-me

Introduction

1. **Please, OBSERVE THE LINKS I provided below because many of them are linked to my CV or website (where I aggregate all my information).**
2. **My CVs are available in the following links:**
   - Printable (2 versions):
     - **Full Stack Developer**
       https://paulojeronimo.com/cv/en/resume-web3.html
     - **DevOps and Site Reliability Engineer**
       https://paulojeronimo.com/cv/en/resume-devops.html
   - Forty (40) pages (better not print it to be eco-friendly! ;-)):
     https://paulojeronimo.com/cv/en/

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1. How do I develop software in "The Artificial Intelligence and Web3 Era"?

I am a computer scientist who appreciates many technologies.

So, Today, I am not in a position to consider myself an expert in any of them.

I work on projects that match my interests at a given time. I like to train my current skills, but in many cases, I prefer to acquire new ones. Frequently, I do this when I decide to switch to a different job.

For example, when I realized I was immersed in Java development and had little infrastructure knowledge, I reduced my work as a developer in this language. I went deeper into the work as someone who needed to work with middleware and, for that, to acquire more infrastructure skills.

Similarly, in 2015 I realized that I had little knowledge of JavaScript. So, I left my job as a back-end developer and started using Node.js more to produce software for both the back-end and the front-end.

In 2017, due to my interest in Blockchain, I immersed myself in developing solutions with Blockchain. Primarily focused on Ethereum.

In 2019, when I returned to Brazil, I decided to specialize a little more in React to produce some personal applications to the wellness area. **More details here: https://finisher.tech/dao**.

In 2020, while working for OSO DevOps (technically led by Sion Smith, partner of a London partner company of AWS and Kafka), I grow up my skills in these two tecnologies.

This year (2023), due to my interest in evolving one of my projects related to Web3 solutions for the wellness area, I decided to broaden my knowledge a little about a language that I also use in projects, but on an occasional basis. I'm developing an application using Python, TypeScript and deciding if I use Solidity or Vyper:

The Phi Treadmill Training Plan App (TPTTPA) - A full-stack wellness project to show you the integrated use of some modern technologies
https://paulojeronimo.com/tpttpa/tpttpa.pdf

In this way, due to the large number of information technologies and programming languages that I need to use to develop, the use of Artificial Intelligence tools (to help me build solutions) is increasingly becoming part of my daily programming life.

Today, in the era when ChatGPT popularized Artificial Intelligence, I practically develop my software solutions daily with its help. I am also experimenting with integrating my favorite *Personal Development Environment* (Neovim) with GitHub Copilot.

2. My history as a Full Stack Developer (related to Java programming)

2.1. One of my current Java projects (working for a company in Brazil)

I'm working on a project for the **municipal IT company in Rio de Janeiro (IplanRio)- https://iplanrio.prefeitura.rio/**.

This project aims to prepare the budget and control expenses for social programs organized by the city hall.

I am happy to work with the technologies I use in this project, as they are modern yet closely related to my professional background related to Java.

In this project, I'm accumulating three (3) roles: Software Architect, Full Stack Developer, and Quality Assurance Automation Engineer (I love to develop solutions based on test specifications). This project's current team is tiny: me (working on the roles above), a back-end dev, a front-end dev, and some people working as requirements analysts and UX.

The two developers on the project were working very independently and separately. I was hired to guide them in their activities, validate their work, and make their work more integrated.

So, currently, I lead the development team in its activities, evaluating and approving their commits on GitLab. Also, I act as a Full Stack Developer and simultaneously ensure the delivery of the entire application to the client's environment.

On July 4, 2023, we delivered our first sprint in this project, publishing the back-end and front-end components in the OpenShift environment of the IplanRio.

In the following weeks, to reduce the problems we encountered in the first delivery of the project, I expect to be responsible for creating an OpenShift environment on AWS and also an installation of GitLab CI/CD to have a development environment equal to IplanRio's. However, out of its territory and managed by me.

We are developing, on the back end, some Quarkus microservices. They access a MariaDB database. Also, it has some internal integrations with APIs. Some microservices may talk with others using Kafka or some similar tech.

Unfortunately, this project is not public (nor open source). Its users are municipal employees. So, to authenticate and authorize them in the application, an integration with a proprietary API developed by the agency itself, called Cerberus, is used. However, maybe some external modules of this application will be visible to Internet users. In this case, Keycloak (RH-SSO) will be used.

In this link is a public and old presentation developed by someone working at IplanRio and for previous projects, which describes some of the technologies used by this company.

In the front end of this project, we are using Next.js and ChakraUI to build the UI.

2.2. As a Java Programmer ...

I have 25 years of development with this language (from 1998 to Today).

My first official Job programming Java was in 1998 when I worked for a company called Politec in Brasília, where I live with my wife since 1997.

In 2003 I left Brasil Telecom because I partnered with Helder da Rocha from Argo Navis to teach his Java and XML courses in various places in Brazil. In this way, I taught many of the training listed in the delivered courses section of my CV for several Brazilian companies.

In March of 2007, we both started to work on a project for Sun Microsystems. This project and my partnership with Argo Navis lasted until the end of 2007. After that, as a consultant, I started teaching JBoss official courses (from RedHat).

At the beginning of 2008, one customer of this course offered me a more stable position as an employee in a middleware specialist role. So, I accepted this job (Ambima). Since then, I have gone through several other jobs where I developed my skills as a programmer, middleware specialist (JBoss, WebSphere, GlassFish, etc.) or software architect focusing on this language.

In 2011, I was so involved with GlassFish, Java EE, and IDEs for Java that Arun Gupta invited me to talk at the Oracle Technology Network Developer Day 2011. I found a citation for my talk by searching for it on Google. You can see many other posts related to this period of my life by accessing the blog Lado Servidor where some friends like Bruno Borges and Marcelo Ancelmo also wrote some articles for it.

I had the opportunity to participate in many Java events. One of my last presential events related to Java was JavaOne 2017 in San Francisco. That time was the launch of the Java 9 version. Even without being a speaker, I enjoyed the event because I saw some Java Rockstars friends with whom I learned a lot. Some of them worked with me on past projects. Here are some of the pictures I took at this JavaOne:

- 1 ← *Bruno Borges and Michael Nascimento*
- 2 ← *Rafael Benevides and Edson Yanaga*
- 3 ← *Bruno Souza and Venkat Subramanian*
- 4 ← *books that I bought*

Before my last job with Java, I worked with it in many other opportunities, including one outside Brazil in 2018, working for Thales in Lisbon programming with the Spring stack.

In 2020 I also build some Java programs using Quarkus.

One of my last jobs with Java was at Fóton Informática in 2021. But, I was working with ancient and outdated versions of Java in this position. So, I chose not to evolve in this work in August 2021 because I was frustrated with this situation. After this job, I decided to look for jobs more focused on other programming languages (like JavaScript, TypeScript, and Solidity) because this was more aligned with the necessary skills that I wanted to acquire.

I've been watching many videos related to the evolution of the Java language (from version 9 to 19) to update my knowledge about this language. But currently (and since 2021), I am more fluent in solving problems using the programming languages I mentioned.

2.3. Covering Java Virtual Machine strengths and weaknesses ...

I have 14 years (from 2008 to 2021). I was responsible, in many cases, for investigating problems like memory leaks in JVM applications developed for customers in several areas. Working for RedHat, and even much before this, I used tools provided by the JVM to analyze memory consumption, performance, and so on.

In 2008, working for Ambima, I was responsible for doing many tasks in the production environment for the Central Bank of Brazil's critical applications (SELIC and Ofpub). Since that, and for some other customers (Algar, IBM, Mais2x), I have done the same task related to tuning and troubleshooting of JVM for some applications in these customers.

3. My history programming with Web3 technologies

I have 7 years working with Blockchain technologies (from 2016 to Today).

In 2016, I understood the actual value of Bitcoin and started investing in it. But, it was **in 2017** that I got hands-on experience with Ethereum. Unfortunately, I'm not much of a video recorder. I prefer to write code or articles and tutorials. But, even without consistency in the publication of videos, since that time, I have some published videos about vídeos Blockchain or Ethereum.

In 2017, when I had some Bitcoins in my wallet, which I put to good use. I sold it at a good moment to buy an apartment for my daughter, who recently got married.

Since this year, I have been very involved with Blockchain doing severeal related things:

- I recorded in this repository a playlist of videos (in Brazilian Portuguese - see the link above) derived from a work done by Anders Brownworth.
- I developed my skills with Ethereum and got involved with its blockchain to understand more deeply how it worked.
- I mentored about it in a Hackathon offered by a startup accelerator in Brasilia when I was already living in Lisbon. At that time (2018), I created my first video playlist about Ethereum.

In this year, while working for Thales in Lisbon, I participated in Mobility Hack Lisbon. Along with a fantastic team, I received a prize of two thousand euros for participating in this Hackathon:

The focus of this Hackathon was the development of solutions involving the use of Blockchain for the automotive area. Mercedes Benz organized it. We created a solution by programming Smart Contracts in Ethereum, *backend* in Kotlin, and *frontend* in React. The objective of this project was to obtain data from the Electronic Control Unit (ECU) of vehicles and publish it on the Ethereum Blockchain to prevent used vehicle sales fraud.

My main objective in going to work in Europe this year was to find a position as a developer in Blockchain. But, I didn't get that position because my resume was focused on my experience as a Java EE architect/ Java developer. It also featured my experience in Linux and products like Docker,

Kubernetes, and RedHat (JBoss, OpenShift, Keycloak). So, as I needed a quick job, when I arrived in Lisbon, I took the first job demand and assumed the position of Software Architect at Thales.

I spent a year working for Thales. Apart from architectural issues, I also developed my skills in DevOps and BigData matters. However, I had to return to Brazil because my wife could not obtain leave from her work to stay by my side.

In 2019, back in Brazil, I started my startup, where my desire (in the long term) (is to build Move-To-Earn DApps. This desire stems from my desire to encourage people in sports practice by offering them new ways to motivate themselves (based on Tokens ERC-20 and NFTs). As an amateur triathlete, who has already finished three (3) Ironman races, I know very well the need and benefits of daily sports practice in people's lives.

But, undertaking alone in Brazil, and in this area that I decided to act, is quite complicated, and success depends on several factors (not just technical ones). Also, we were not yet in a time when one-person companies succeeded through Artificial Intelligence. So, while perfecting myself in Web3 and developing ideas and prototypes with technologies, I looked for partners who would complement me in non-technical areas. In parallel, I joined programs to validate my thoughts and get investments.

In 2020 I offered (free) webinars on programming DApps with React and Solidity to demonstrate my technical knowledge about these subjects.

In 2022, a company I had already worked for, ilia, asked me to mentor in Web3. I trained several programmers in technologies related to Ethereum, Polygon, and NEAR. In the latter, I started to learn Rust for developing Smart Contracts in this language.

That year I also participated in several events in the Web3 area, such as Blockchain Rio and Ethereum Hackathon in São Paulo. I also started a DAO (called Finishers.DAO) related to Web3 projects I have ideas for.

Please revisit this page to read more about.

Today, I am also using the power of Web3 technologies to build a platform for building and selling tutorials and ebooks (only for my use initially). After this platform works, I intend to evolve it for something bigger and share its use with other technical writers.

4. My experience regarding ...

4.1. ... distributed applications

Regarding Distributed Systems, I have 21 years (from 2002 to Today)

My specialization at the University of Brasília, in 2002, was on "Distributed Systems and Object Orientation". At that time I started to get involved a lot with middleware like Tomcat and JBoss.

Starting in 2014, when I spoke at a Red Hat event about Docker, I also started following projects that were emerging like OpenShift, an implementation of Kubernetes.

I also worked with the development of microservices using Cloud Foundry. In the Cloud Computing world, distributed systems are the law and some of my experiences with AWS, Azure and GCP have reinforced my view.

> Again, check the links above. In there I give some details of my using of these technologies.

4.2. … other programming languages (beyond Java)

At the beginning of my career as a programmer, I worked with languages such as COBOL (in 1993), Pascal, C, and C++ (from 1993 to 2001).

Today, I solve problems by coding software in many programming languages:

- Bash
- Groovy
- JavaScript
- Kotlin
- Python
- Rust
- Solidity
- Vyper
- TypeScript

> Again, check the links above. In there I give some details of my using of these programming languages.

4.3. … developing simple, modular, testable, and high-quality code

I like the concept of Java 9 Modules and I presented it in a live video recorded for DFJUG, in 2017.

Also, I like the concept of ES6 modules and the way that NPM packages works. I found this out when I learned about creating my own packages in JavaScript.

I'm Test Infected since I met the JUnit creator (Kent Beck), in person, when he comes to the event Extreme Programming Brasil, in 2002 (*see some links about this on my HOME page*).

I delivered many courses about this Testing in Java (using TDD or BDD), for many companies in Brazil. So, I have experience in Unit Testing, Integration Testing, Acceptance Testing, and Performance Testing (with JMeter, ab, etc.).

4.4. … test-driven development processes and continuous integration tools

Regarding testing, I believe that I already give a response to this question.

My first contact with a continuous integration (CI) tool was when working for Ambima (in 2008). Today I have worked with projects running in a lot of CI tools like Jenkins, GitLab CI/CD, Travis CI, CircleCI, GitHub Actions, and AWS CodePipeline. I'm also capable to install and configure pipelines on many of this tools.

4.5. … developing libraries and platforms for use and deployment by application developers

For many companies that I worked for, since I my first jobs which Java, I had to create libraries and frameworks to facilitate the live of other developers. Unfortunately, I can’t prove this with code that I have published on my GitHub account because my work on this subject was did in private repositories of source code in that companies.

4.6. … systems programming, including providing services to other software

Related to providing operating systems, computational science applications, industrial automation, and SaaS applications, my main areas of work in the development area were for large companies in the banking or telecommunications area. Due to the speed requirements necessary for the functionalities offered by applications in these areas, I got very involved with code optimization and the application's relationship with the operating system's resources.

I participated in the maintenance and production of B2B applications, with a lot of integration between systems through market standards.

My experience with Java middleware (Tomcat, JBoss, Keycloak, etc.) has also given me excellent insight into the lower-level features of this software and their integration with operating systems, networks, databases, etc.

5. About this document

5.1. It is written in the same way as I code (Docs-As-Code)

I create documents like I code, and I like to build them this way. My website, CVs, and even this document were produced in what it's called Docs-As-Code. My papers are versioned and stored in a private or public Git repository. In many of my documents (like in my CVs) I use Conditional Preprocessor Directives to create different versions of the same document. This is a very nice feature provided by Asciidoctor.

5.2. It is written with my favorite text editor (Neovim)

Neovim is my favorite text editor. I use it to write all my documents (including this one) in the AsciiDoc format. Regarding how I use this editor, I am also creating a tutorial about it:

Configuring Neovim as a Personal Development Environment (PDE) to Full Stack Development
https://paulojeronimo.com/tutorials/neovim.pdf

5.3. It is built with some lovely tools (Bash, Asciidoctor)

In many cases, I also create a Bash script to automate some tasks related to building (with Asciidoctor) or sharing the document. Related to this subject, you can read, as a example, how do I build my own CV.

5.4. It is written like I write my tutorials (to sell them)

In one of my last tutorials (published on YouTube in Portuguese), I also started to talk about this subject:

How would a software developer create technical documents and sell them using the disruptive power of Web3 technologies? (Part I)
https://paulojeronimo.com/asciidoctor-web3-article/private/sample.pdf

5.5. People mentioned

- Anders Brownworth - https://www.linkedin.com/in/andersbrownworth/
- Arun Gupta - https://www.linkedin.com/in/arunpgupta/
- Bruno Borges - https://www.linkedin.com/in/brunocborges/
- Bruno Souza - https://www.linkedin.com/in/brjavaman/
- Edson Yanaga - https://www.linkedin.com/in/yanaga/
- Helder da Rocha - https://www.linkedin.com/in/helderdarocha/
- Kent Beck - https://www.linkedin.com/in/kentbeck/
- Marcelo Ancelmo - https://www.linkedin.com/in/marceloancelmo/
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- Sion Smith - https://www.linkedin.com/in/sionsmith/
- Venkat Subramanian - https://www.linkedin.com/in/vsubramaniam/